



United International University

Department of Computer Science and Engineering

SOC 2101: Society, Environment and Engineering Ethics

Final Exam. : Fall 2024

Total Marks: 25 Time: 1 hour 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the 5 questions. Numbers to the right of the questions denote their marks.

1. Mr. X, after years of hard work and dedication, was promoted to the position of "Software Lead" in his company. During the promotion ceremony, he received praise and admiration from his colleagues. However, after taking over the new role, he began to exhibit behaviors that negatively impacted the company. Driven by his ambition to achieve extraordinary profits within two quarters, Mr. X implemented several extreme measures. He installed cameras in the offices of junior developers, believing it would improve efficiency. Additionally, he took on projects outside the company's expertise, such as three blockchain projects, despite having no blockchain developers. He assumed that the existing engineers could manage by relying solely on documentation. When the engineers requested funding for an online blockchain course, he denied it, deeming it an unnecessary expense. During project meetings, Mr. X often bragged about his experience and dismissed remarks from junior developers, even though they worked directly with the codebase. As project deadlines approached, he enforced unpaid overtime on his engineers. Over time, the company's performance declined, and eventually, Mr. X was dismissed due to his mismanagement. Analyze the ethical violations committed by Mr. X in light of the ACM Code of Ethics. [5]
2. A financial institution is developing a machine learning algorithm to predict whether applicants will repay their loans based on their credit history, income, and job details. The algorithm is meant to make the loan approval process faster and safer. However, there are concerns that the model might unfairly reject people living in certain neighborhoods or working in industries with unstable incomes, even if they are capable of repaying loans.
 - i Among the following three biases – cognitive, statistical, and historical – which type of bias is most likely to occur in this situation? [1]
 - ii What ethical issues could arise from using such a system? Discuss how algorithmic bias in the model might create unequal and unfair access to loans? [2]
 - iii Suggest ways to mitigate bias and make the system fairer so it doesn't unfairly disadvantage certain applicants. [2]
3. In an ambitious initiative called Project E.L.ON (Emulated Life: Ongoing Network), Twitter (now X) developed an AI-driven chatbot designed to mimic the personalities of deceased loved ones using their past tweets, direct messages, and voice recordings from Spaces. Without explicit user consent, the AI was trained on years of publicly available and privately archived digital interactions. In October 2024, Twitter (now X) quietly launched a beta test for a select group of users who had recently experienced loss, offering them AI-generated conversations that felt eerily authentic. Many users found comfort in reconnecting with digital echoes of their loved ones, while others felt deeply unsettled. The company later presented Project E.L.ON as a breakthrough in AI-driven emotional support, claiming it helped users process grief.

What ethical dilemmas does Project E.L.ON raise, and what safeguards should have been implemented before conducting such a project? [5]
4. A retail company wants to improve its customer service and sales by analyzing shopping habits. They install smart cameras in their stores that use facial recognition to track customers' movements, time spent in different sections, and emotional reactions to products. The company does not explicitly inform customers about this data collection, assuming that since it is done inside their store, consent is implied.

Is it ethical for the company to use facial recognition without explicit customer consent? What steps could have been taken to ensure transparency and customer privacy? Write down the data collection ethics with a consent letter. [5]

5. In Demon Slayer, Nichirin swords have always been the go-to weapon against demons. But with stronger threats emerging, Tengen Uzui has introduced some flashy upgrades.

He first developed a built-in alert system for Nichirin swords. If a High Upper Moon appears, the sword instantly signals the Hashiras and nearby slayers for backup. He also created artificial sunlight Nichirin weapons, powered by burning coal to generate UV light. While they are effective, they produce tons of waste and require constant fuel.

Some argue that this technology makes slayers too dependent on tools instead of mastering Breathing Techniques, and it harms the environment. Others believe it enhances their power, giving them an edge against stronger demons.

Would you stick to tradition or embrace Uzui's high-tech revolution? As a technologist, how would you refine his inventions to make them more efficient?

[5]